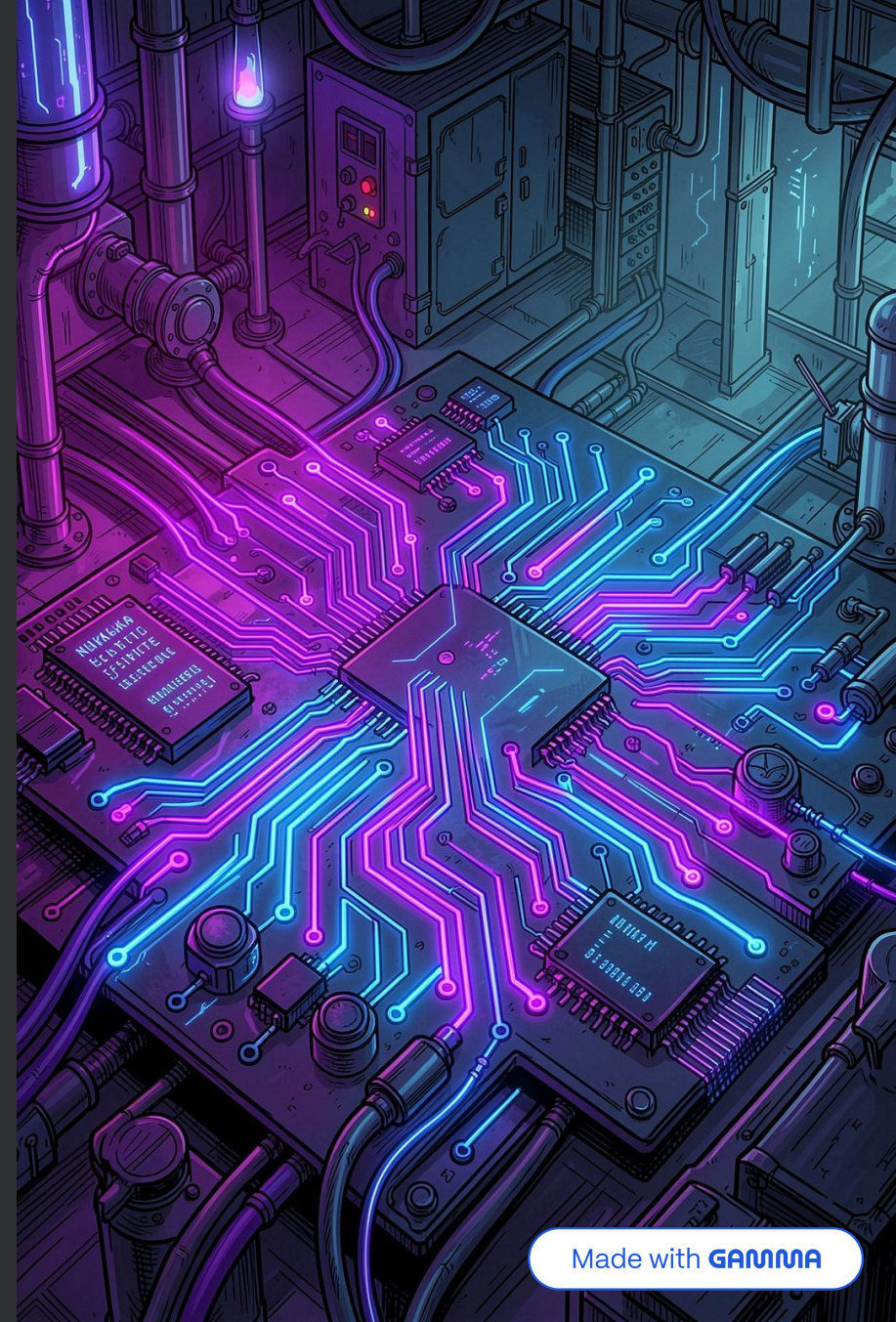


CircuitMate ⚡

An AI-powered electronics troubleshooting assistant — converting circuit symptoms and images into structured fault diagnosis, step-by-step guidance, and engineering calculations. Built for students, hobbyists, and hardware developers.

AI ELECTRONICS ASSISTANT



Made with GAMMA

The Problem

Debugging Complexity

Pinpointing why a circuit fails — blown transistors, wrong resistors, misconfigured timers — is tedious and frustrating.

Calculation Overhead

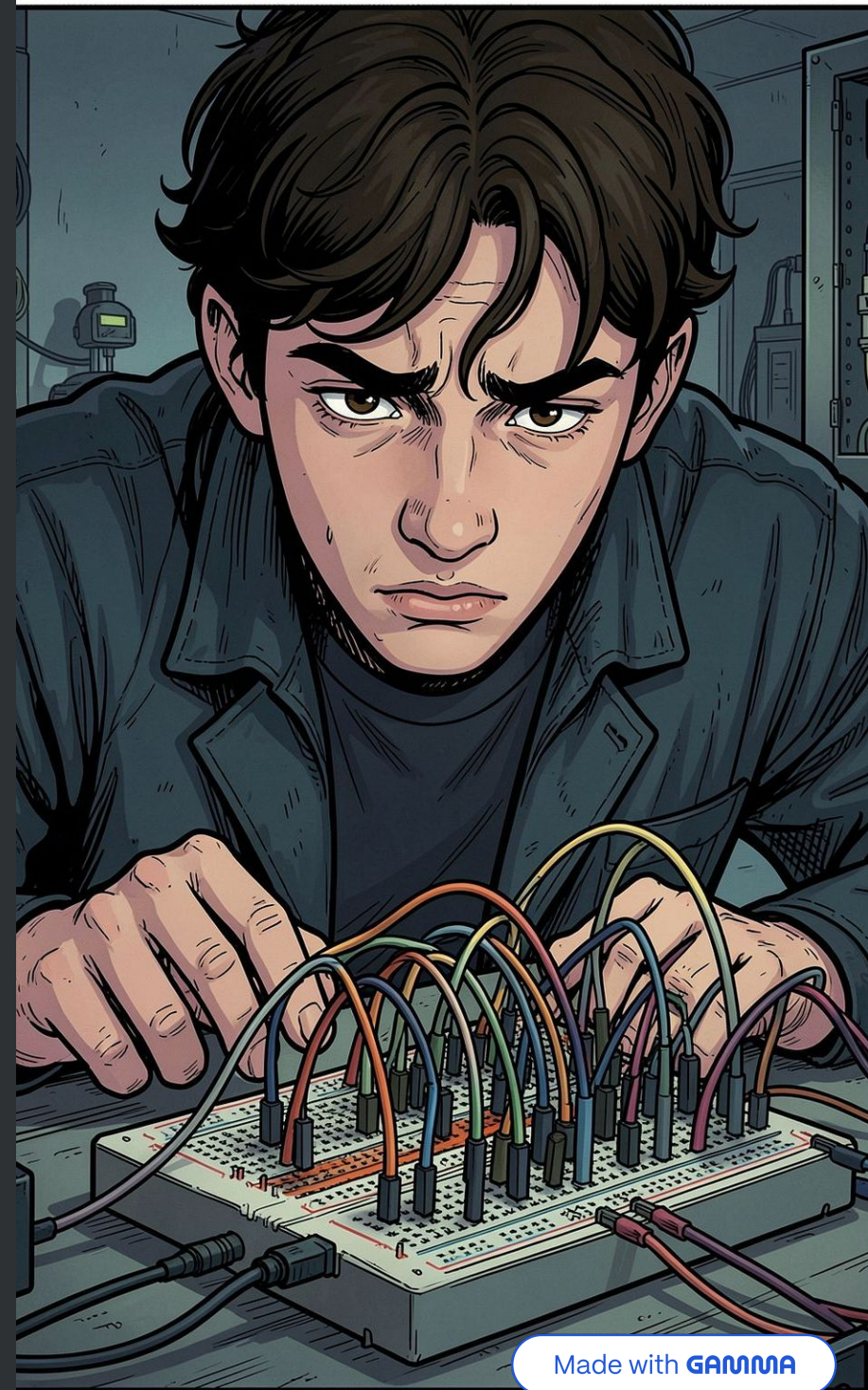
Switching between formulas for voltage dividers, RC filters, and LED limits slows design work.

Documentation Friction

Writing lab reports and organizing Bill of Materials takes significant manual effort.

No Visual Validation

Standard AI assistants can't interpret breadboard layouts or provide structured schematic representations.



AI Circuit Troubleshooter

How It Works

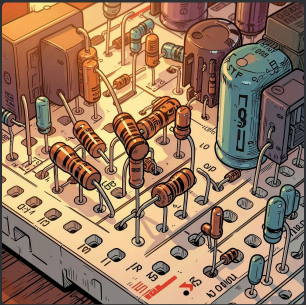
Describe your circuit problem in plain language — CircuitMate analyzes it and returns a structured diagnosis.

📄 **Example:** "Arduino + LDR + LED. LED stays ON even when it's dark." → AI identifies possible causes and generates a troubleshooting checklist.

Responses include component identification, connection explanation, working principle, possible faults, and improvement suggestions.

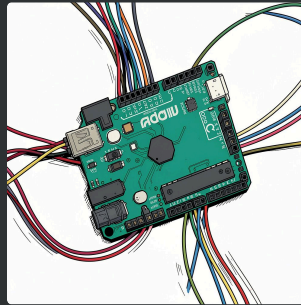
Circuit Image Analyzer

Upload a photo of your breadboard or handwritten schematic — CircuitMate identifies components and explains the visible circuit.



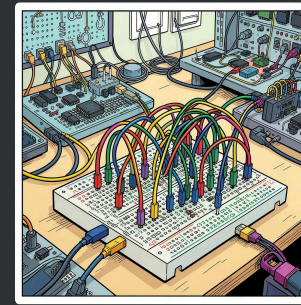
Passive Components

Resistors, capacitors, LEDs identified by type and value.



Microcontrollers

Arduino, ESP boards, and ICs detected and labeled.

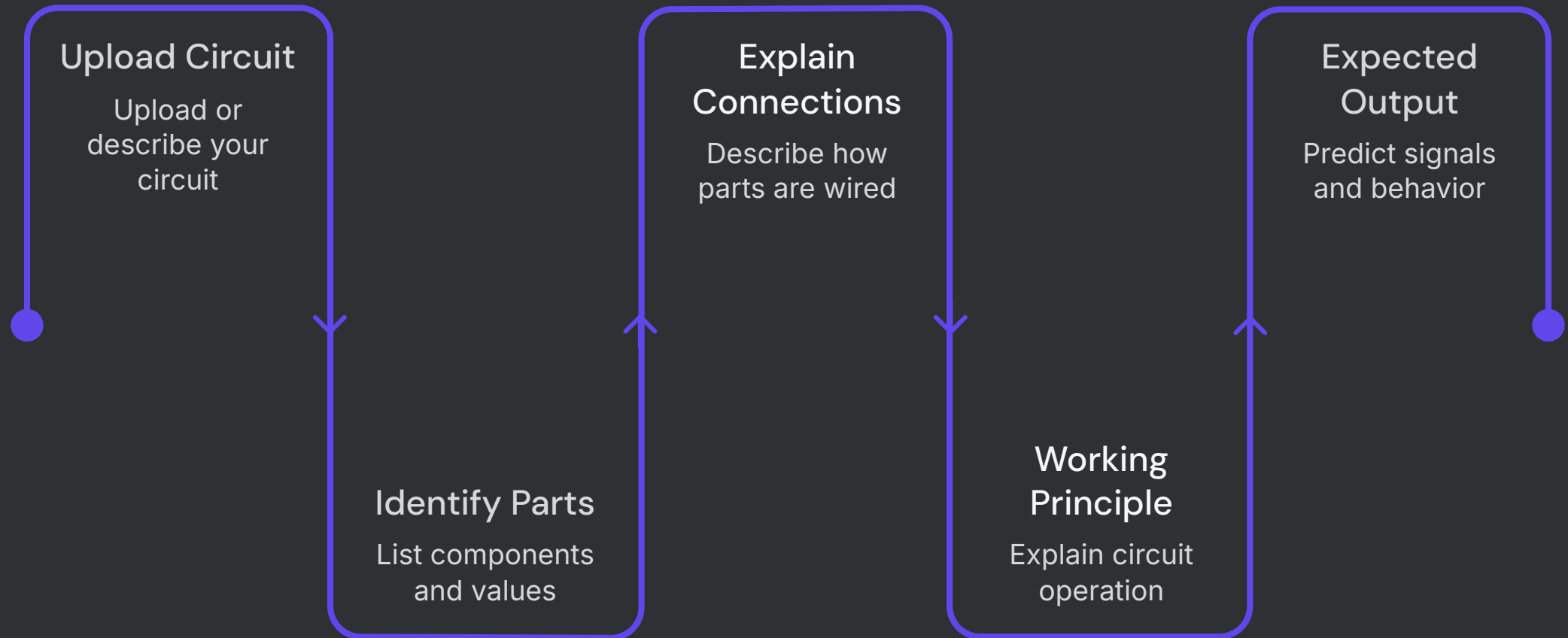


Wiring & Connections

Breadboard connections mapped and explained.

Explain My Circuit

Upload or describe any circuit and receive a complete breakdown — ideal for first-year ECE students.

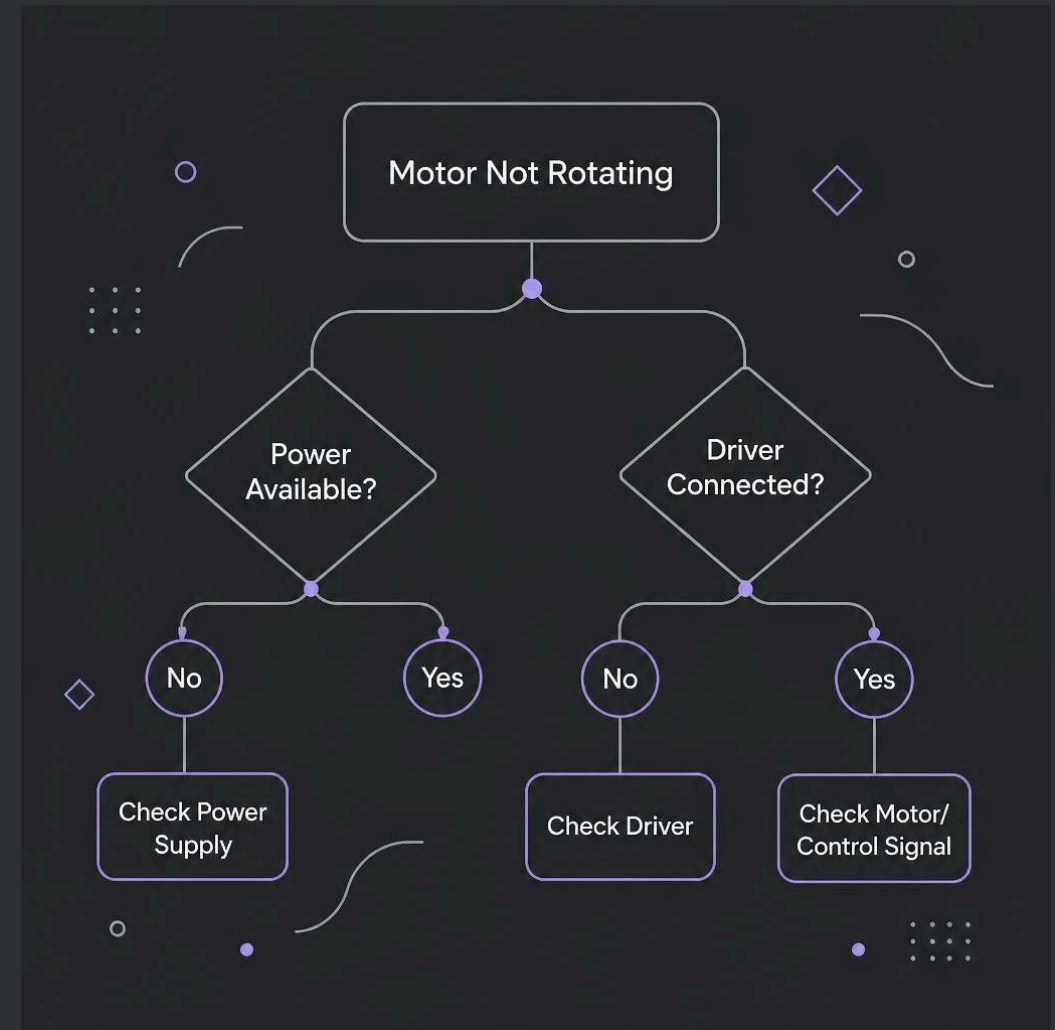


This structured pipeline helps students understand not just *what* a circuit does, but *why* — building real comprehension over rote answers.

Fault Finder

Symptom-Driven Diagnosis

Give CircuitMate a symptom like **"Motor is not rotating"** and it generates an interactive troubleshooting tree.



Is power available?

No → Check power supply

Is driver connected?

No → Check driver wiring

Check motor & control signal

Verify connections and replace if needed

Electronics Calculator Toolkit

Built-in calculators eliminate formula-switching and keep you in the flow of design.



Ohm's Law

Voltage, current, and resistance calculations.



LED Resistor

Safe resistance and power dissipation.



Voltage Divider

Output voltage ratio computation.



RC Filter

Cutoff frequency and time constant.



555 Timer

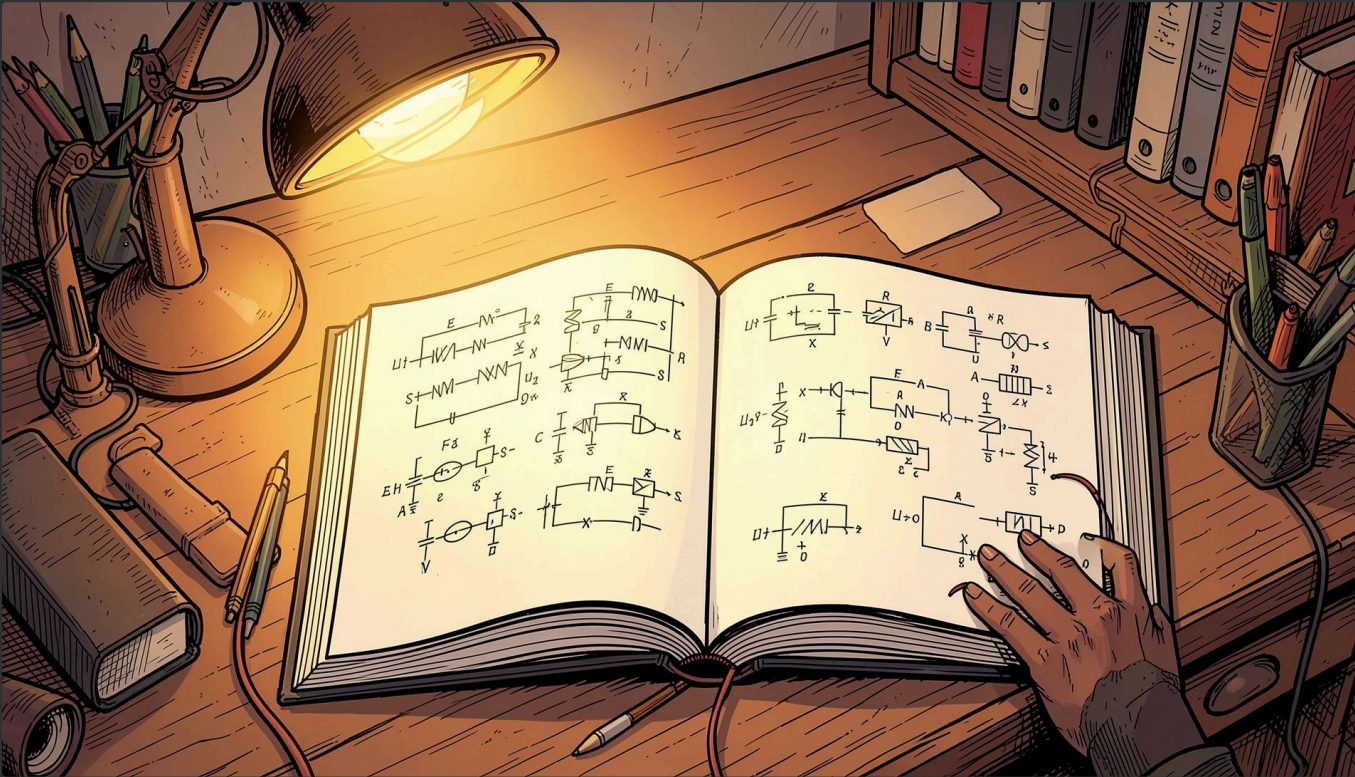
Astable/monostable frequency and duty cycle.



Capacitor Networks

Series and parallel combinations.

Learning Mode & Component Library



Learning Mode

Instead of just giving answers, CircuitMate explains: **Component** → **Function** → **Symbol** → **Working** → **Applications** → **Common Mistakes**. Students learn the *why* behind every answer.

Component Library

Search any part (e.g., **BC547**) and instantly see pin configuration, typical applications, key specs, and common mistakes — a pocket reference built in.

Tech Stack & Hackathon MVP

Technology Stack

Frontend

HTML5, Tailwind CSS,
Vanilla JS / React

AI

Google Gemini API +
offline fallback / IBM
watsonx

Backend

Python, Flask / FastAPI,
SQLite

Browser APIs

Web Speech, File API,
LocalStorage

MVP — Build These First

1. Describe electronics problem
2. AI diagnosis
3. Step-by-step troubleshooting
4. Circuit/component explanation
5. Ohm's law + basic calculator

Add image analysis and advanced features once the core system works.